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To: DoD Identity Protection and Management Senior Coordinating Group (IPMSCG)
From: Mr. Michael Butler, Deputy for Identity

RE: Updated Technical Bulletin—Implementation of SSN Removal from DoD ID Card Barcodes

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1. Introduction

This Technical Bulletin supersedes the previous version released on December 20, 2011. As part of the Department's Social Security Number (SSN) reduction activity (i.e., Under Secretary of Defense for Personnel and Readiness Memorandum, "Updated Plan for the Removal of Social Security Numbers (SSNs) from Department of Defense (DoD) Identification (ID) Cards," 5 November 2010), DoD will remove the SSN from barcodes that are printed on the Common Access Card (CAC) and the remaining family of DoD identification cards. DoD completed the initial phases of the transition by removing the SSN printed on the printed surface. The next phase will remove the SSN from the Code 39 barcodes. DoD updated implementation will:

- Continue with effort to remove the SSN from Code 39 barcodes by the end of **December 2012**, and
- Delay the removal of the SSN from PDF 417 barcodes **till the end of December 2013**. A separate technical bulletin will be released at a later time to specifically address this part of the Department's implementation.

The purpose of this document is to describe how the Department will implement the Code 39 barcode changes.

2. Audience & Scope

DMDC has written this document for the community of users, application developers and system maintainers who interact with the barcodes on DoD ID cards. This document will outline the future format of the Code 39 barcodes on all DoD ID cards (CACs and Teslin).

3. Description of the Changes

Removal of SSN: DMDC considered completely removing the SSN field from the structure of the Code 39 barcodes and realigning the remaining fields, however, discarded this approach due to the extensive impact that it would have on all applications that read and interpret the individual fields. DMDC decided

that the best approach (i.e. least impact) for removing the SSN is to keep the current structure and repurpose the field. The SSN field will become a unique card security identifier that distinguishes each card from any other card issued to an individual. There is currently a random character (characters 0-9 and A-Z) card instance identifier in the Code 39 barcode, but the identifier is not guaranteed to be unique across all ID cards issued to an individual throughout their association with the DoD. DMDC designed the new format for the SSN field to address this ambiguity. The new field format will allow Physical Access Control Systems (PACS) that keep a history of an individual's enrolled cards, such as the Defense Biometrics Identification System (DBIDS), to more easily distinguish all previously issued cards from the current card.

DMDC's implementation will set the card security identifier's leading three digits to nines to serve as a flag for an invalid SSN number, since the Social Security Administration does not issue valid SSNs that begin with "999". The remaining six digits provide 1,000,000 unique numbers to identify cards issued to each individual. The SSN field in the Code 39 barcode will continue to be encoded as six base-32 characters.

SSN Format: 999-XX-XXXX
Example: 999-55-1212 (base-10) → TP7S7C (base-32)

Applications using the SSN within the barcode should already be aware of the migration to EDIPI (instead of the SSN) as the unique identifier for individuals within the DoD. System maintainers are expected to modify their systems to use the EDIPI instead of the SSN.

Person Designator Type Code: DMDC will be adding a new Person Designator Type to "X" to the barcode when the Person Identifier is the random number to differentiate from the SSN.

Outlined in Appendix A and B are the new barcode formats for CAC/Teslin linear barcodes and CAC 2-diminsional barcodes. To support application testing and development, samples of the changes are provided in Appendix B.

4. Timeframe

DMDC will retain the current structure of the Code 39 barcodes, but begin to use the "SSN" field as a six digit unique identifier with the first three digits set to "999". Maintaining the existing bar code structure is expected to minimize the impact on software applications, system vendors and the bar code using community. These changes will be implemented on newly issued DoD identification cards beginning in **December 2012**.

5. Questions and Technical References

Technical details about the barcode implementation that includes these changes can be found within DoD ID Card Bar Code and Software Developer's Kit at https://www.dmdc.osd.mil/smartcard/owa/ShowPage?p=BARCODE .

Specific questions or technical inquiries should be sent to barcodesupport@osd.pentagon.mil.

Appendix A: CAC and Sponsor Teslin Code 39 Barcode Format

The CAC Code 39 barcode layout contains seven data elements. It follows the existing layout for civilian ID cards as much as possible. Formats and field descriptions are shown in Figure 1 and Table 1, respectively. Instructions for decompressing the Person Designator Identifier and DoD EDI Person Identifier fields are provided in the DoD ID Card Bar Codes & Software Development Kit, version 7.4.0.

Figure 1 CAC Code 39 Barcode Format

VC	Card Security Identifier						PT	DoD EDI Person Identifier							PC	BC	CI
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18

VC = Barcode Version Code
PT =Person Designator Type Code
PC = Personnel Category Code
BC = Branch Code
CI = Card Instance Identifier

Table 1 CAC Code 39 Barcode Detail

Field	Length	Description
Barcode Version Code	1	"1" - Indicates format version and card type for the bar code
Card Security Identifier (CSI)	6	The six-digit CSI, in the range 00-0000 to 99-9999, is prepended with "999' to make a nine-digit number and stored as a six-digit base-32 number in the corresponding range 000000 to TPLIFV.
Person Designator Type Code	X	See Appendix C, DoD ID Card Bar Codes & Software Development Kit, version 7.4.0
DoD EDI Person Identifier (EDIPI)	7	DEERS-assigned numeric identifier (compressed). The ten-digit EDIPI in the range 0000000000 to 9999999999 is stored as a seven-digit base-32 number in the corresponding range 0000000 to 9A0NOVV.
Personnel Category	1	See Appendix C, DoD ID Card Bar Codes & Software Development Kit, version 7.4.0
Branch Code (Service)	1	
Card Instance Identifier	1	A machine-generated random code that is used for security purposes to distinguish a card from other cards that have been issued to a cardholder. Formerly, Security Code.
Total Characters	**18**	

Appedix B: Sample Barcodes

Type of DoD ID Card	Sample Barcode	Scanned Data (Raw)	Decoded Data
One-Dimensional Code 39 Barcode for CAC and Sponsor Teslin			
CAC/Teslin Code 39 V1		1TOQQG0X14PC0MIAA5	Bar Code Version Code: 1 Person Designator Identifier: 999123456 Person Designator Type Code: S DoD EDI Person Identifier: 1234567890 Personnel Category Code: A Branch Code: A Card Instance Identifier: 5
CAC/Teslin Code 39 V1		1TPBOMMX10DINPAEDL	Bar Code Version Code: 1 Person Designator Identifier: 999678678 Person Designator Type Code: S DoD EDI Person Identifier: 1087987498 Personnel Category Code: E Branch Code: D Card Instance Identifier: L
CAC/Teslin Code 39 V1		1TP89BEX102M4HMHMB	Bar Code Version Code: 1 Person Designator Identifier: 999564654 Person Designator Type Code: S DoD EDI Person Identifier: 1076564534 Personnel Category Code: H Branch Code: M Card Instance Identifier: B

CAC/Teslin Code 39 V1		1TP1GB0X0USQNB6AC3	Bar Code Version Code: 1 Person Designator Identifier: 999342432 Person Designator Type Code: X DoD EDI Person Identifier: 1036868966 Personnel Category Code: A Branch Code: C Card Instance Identifier: 3
CAC/Teslin Code 39 V1		1TP5GN6X104TQJLFF5	Bar Code Version Code: 1 Person Designator Identifier: 999473894 Person Designator Type Code: X DoD EDI Person Identifier: 1078913653 Personnel Category Code: F Branch Code: F Card Instance Identifier: 5
CAC/Teslin Code 39 V1		1TP1IVEX0V56K9HR16	Bar Code Version Code: 1 Person Designator Identifier: 999345134 Person Designator Type Code: X DoD EDI Person Identifier: 1045647665 Personnel Category Code: R Branch Code: 1 Card Instance Identifier: 6
CAC/Teslin Code 39 V1		1TPL3H5X10D2PO7LNO	Bar Code Version Code: 1 Person Designator Identifier: 999984677 Person Designator Type Code: X DoD EDI Person Identifier: 1087465223 Personnel Category Code: L

			Branch Code: N Card Instance Identifier: O
CAC/Teslin Code 39 V1		1TPGJV1X0UQACRIQNR	Bar Code Version Code: 1 Person Designator Identifier: 999837665 Person Designator Type Code: X DoD EDI Person Identifier: 1034236786 Personnel Category Code: Q Branch Code: N Card Instance Identifier: R
CAC/Teslin Code 39 V1		1TP4T2TX0V21OKSKD2	Bar Code Version Code: 1 Person Designator Identifier: 999453789 Person Designator Type Code: X DoD EDI Person Identifer: 1042342556 Personnel Category Code: K Branch Code: D Card Instance Identifier: 2
CAC/Teslin Code 39 V1		1TOU6V8X10F8CA7MX9	Person Designator Identifier: 999234536 Person Designator Type Code: X DoD EDI Person Identifer: 1089745223 Personnel Category Code: M Branch Code: X Card Instance Identifier: 9